

D833

Cybersecurity and Information Assurance Capstone

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

You have finished a 40-page technical report on network segmentation weaknesses at a credit union with 22 branches. The CEO, who has a finance background, has five minutes to read your executive summary. What should it lead with?

- A. Network diagrams and VLAN configuration details.
- B. The business risk, the recommended action, its cost, and the decision needed.
- C. A step-by-step account of how the assessment was done.
- D. No cost figures, since executives should never see numbers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A section of your capstone contains firewall rule syntax, test logs, and configuration baselines. Where does it belong?

- A. In the cover letter
- B. In the executive summary
- C. In the technical report or its appendix
- D. Nowhere; technical detail is never needed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

In your scenario, a manufacturer with plants in three states had ransomware on its office network; the plant-floor control network was not affected. You have six weeks and a \$250,000 cap for the proposal's first phase. What should you define first?

- A. Include all networks, since everything must always be in scope.
- B. The scope, constraints, and assumptions, including what is out of scope for this phase.
- C. A list of every possible control for every network.
- D. Which vendor to buy from.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A clinic with 45 staff that just adopted a new electronic health record system is assessed against a framework. Current state: no asset inventory, and MFA on email only. Target: a complete asset inventory and MFA on all remote access. What is the output of a gap analysis?

- A. A list of the target controls only.
- B. A certificate of compliance.
- C. A prioritized list of differences between the current and target states.
- D. An inventory of controls already in place.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Four findings are scored on 1–5 scales. A: likelihood 4, impact 3. B: likelihood 2, impact 5; the CIO raised it in a board meeting. C: likelihood 5, impact 2. D: likelihood 3, impact 5. Using risk = likelihood × impact, which should be addressed first?

- A. D
- B. All four at once, since every finding must always be fixed together.
- C. C
- D. B

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A finding has an impact of 4 and a risk score of 12 using risk = likelihood × impact. What is its likelihood?

- A. 4
- B. 8
- C. 3
- D. 48

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Your proposal for a school district with 18 schools recommends a system that collects logs and alerts staff to unusual logins. Which NIST CSF function does this mainly support?

- A. Detect
- B. Protect
- C. Respond
- D. Identify

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which recommendation maps mainly to the NIST CSF Recover function?

- A. Deploying intrusion detection sensors
- B. Rolling out MFA for all staff
- C. Building a complete asset inventory
- D. Regularly tested restoration of backups, with a plan for communicating during recovery

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Option A costs \$40,000 upfront plus \$15,000 a year in maintenance. Option B is a \$30,000-a-year subscription from a vendor headquartered overseas with 24/7 support. Over three years, which costs less, and by how much?

- A. Option A, by \$35,000
- B. Option B, by \$25,000
- C. Option B, since subscriptions are always cheaper
- D. Option A, by \$5,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Option A costs \$60,000 upfront plus \$10,000 a year. Option B costs \$25,000 a year. After how many years are the total costs equal?

- A. About 1.7 years
- B. 4 years
- C. 6 years
- D. 2.4 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

The board of a 900-bed hospital approved your recommendation to require MFA for all 6,000 staff. The hospital recently won a design award for its lobby. What is the best implementation approach?

- A. Pilot with one department, track specific metrics such as enrollment rate by a set date, then expand in phases with training.
- B. Turn it on for everyone at once, since a single cutover is always fastest.
- C. Skip metrics, since success will be obvious.
- D. Schedule the rollout around the lobby award ceremony.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Your capstone report for a fictional utility with 1.5 million customers cites a popular blog post for a key breach-cost statistic. A peer reviewer flags the source and also suggests switching to a serif font. What should you revise first?

- A. Remove the citation and keep the statistic.
- B. Keep the blog because it has many readers.
- C. Change the font first.
- D. Replace the blog with an authoritative, current source and cite it properly.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

You have finished a 40-page technical report on network segmentation weaknesses at a credit union ~~with 22 branches~~. The CEO, ~~who has a finance background~~, has five minutes to read your executive summary. What should it lead with?

Target word: content of the executive summary **Decides:** An executive reader with five minutes.

A	Adjacent category	That belongs in the technical report.
B	Correct	Executives need what is at stake and what to decide.
C	Confusion set	Methodology is for the technical report.
D	Absolute language	Cost is central to an executive decision.

2. Answer C Concept 1 • transfer

A section of your capstone contains firewall rule syntax, test logs, and configuration baselines. Where does it belong?

Target word: reverse: from content to document **Decides:** Detailed technical evidence.

A	Adjacent category	A cover letter introduces the work; it does not hold evidence.
B	Inverted operation	Executives need decisions, not syntax.
C	Correct	Technical readers need the detail and traceability.
D	Absolute language	Implementers and auditors rely on it.

3. Answer B Concept 2

In your scenario, a manufacturer ~~with plants in three states~~ had ransomware on its office network; the plant-floor control network was not affected. You have six weeks and a \$250,000 cap for the proposal's first phase. What should you define first?

Target word: first step of the proposal **Decides:** A budget, a deadline, and an affected vs. unaffected network.

A	Absolute language	Scope is set by the problem and constraints.
B	Correct	Scope and constraints shape every later recommendation.
C	Dropped element	Drops the budget and time limits.
D	Inverted operation	Vendor choice comes after the problem is defined.

4. Answer C Concept 3

A clinic with 45 staff ~~that just adopted a new electronic health record system~~ is assessed against a framework. Current state: no asset inventory, and MFA on email only. Target: a complete asset inventory and MFA on all remote access. What is the output of a gap analysis?

Target word: what a gap analysis produces **Decides:** Compare current to target.

A	Dropped element	Drops the current state, so no gaps are shown.
B	Confusion set	Gap analysis identifies shortfalls; it does not certify.
C	Correct	The gaps drive the remediation plan.
D	Adjacent category	That is the current state, not the gap.

5. Answer A Concept 4

Four findings are scored on 1–5 scales. A: likelihood 4, impact 3. B: likelihood 2, impact 5; ~~the CIO raised it in a board meeting.~~ C: likelihood 5, impact 2. D: likelihood 3, impact 5. Using risk = likelihood × impact, which should be addressed first?

Target word: top priority **Decides:** Scores: A 12, B 10, C 10, D 15.

A	Correct	$3 \times 5 = 15$, the highest score.
B	Absolute language	Limited resources require ranking.
C	Dropped element	Highest likelihood, but ignoring impact; its score is 10.
D	Camouflage	Who raised it does not change its score of 10.

6. Answer C Concept 4 • transfer

A finding has an impact of 4 and a risk score of 12 using risk = likelihood × impact. What is its likelihood?

Target word: solve for likelihood **Decides:** $12 \div 4$.

A	Wrong stem number	The impact rating.
B	Adjacent category	Uses an additive model ($12 - 4$).
C	Correct	$12 \div 4 = 3$.
D	Inverted operation	Multiplies instead of dividing.

7. Answer A Concept 5

Your proposal for a school district ~~with 18 schools~~ recommends a system that collects logs and alerts staff to unusual logins. Which NIST CSF function does this mainly support?

Target word: CSF function **Decides:** Spotting events as they happen.

A	Correct	Monitoring and alerting find events.
B	Confusion set	Protect prevents; alerts do not block logins.
C	Adjacent category	Respond acts after detection.
D	Confusion set	Identify covers assets and risk understanding.

8. Answer D Concept 5 • transfer

Which recommendation maps mainly to the NIST CSF Recover function?

Target word: reverse: from function to recommendation **Decides:** Restoring operations after an incident.

A	Adjacent category	A Detect control.
B	Confusion set	A Protect control.
C	Confusion set	An Identify activity.
D	Correct	Recover restores capabilities and services.

9. Answer D Concept 6

Option A costs \$40,000 upfront plus \$15,000 a year in maintenance. Option B is a \$30,000-a-year subscription from a vendor headquartered overseas with 24/7 support. Over three years, which costs less, and by how much?

Target word: three-year TCO comparison **Decides:** A = \$85,000; B = \$90,000.

A	Wrong stem number	Counts only one year of A's maintenance.
B	Dropped element	Compares the first year only (\$55,000 vs. \$30,000).
C	Absolute language	It depends on the totals over the period.
D	Correct	$\$40,000 + 3 \times \$15,000 = \$85,000$ vs. $3 \times \$30,000 = \$90,000$.

10. Answer B Concept 6 · transfer

Option A costs \$60,000 upfront plus \$10,000 a year. Option B costs \$25,000 a year. After how many years are the total costs equal?

Target word: solve for break-even year **Decides:** $60,000 + 10,000n = 25,000n$.

A	Inverted operation	Adds the yearly costs instead of subtracting.
B	Correct	$60,000 \div 15,000 = 4$.
C	Confusion set	$60,000 \div 10,000$ uses A's yearly cost as the gap.
D	Dropped element	$60,000 \div 25,000$ ignores A's yearly cost.

11. Answer A Concept 7

The board of a 900-bed hospital approved your recommendation to require MFA for all 6,000 staff. The hospital recently won a design award for its lobby. What is the best implementation approach?

Target word: rollout approach **Decides:** Large user population and clinical operations.

A	Correct	Phasing limits disruption and shows measurable progress.
B	Absolute language	Big-bang rollouts risk locking out clinicians.
C	Dropped element	Without metrics, progress cannot be shown to the board.
D	Camouflage	The award does not decide the approach.

12. Answer D Concept 8

Your capstone report for a fictional utility with 1.5 million customers cites a popular blog post for a key breach-cost statistic. A peer reviewer flags the source and also suggests switching to a serif font. What should you revise first?

Target word: priority revision **Decides:** A key claim rests on a weak source.

A	Dropped element	An uncited claim is weaker still.
B	Confusion set	Popularity is not authority.
C	Camouflage	Cosmetic feedback does not decide credibility.
D	Correct	Substantive evidence problems come before style.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Executive summary vs. technical report	The executive summary leads with business risk, the recommendation, cost, and the decision needed; configurations, logs, and methods belong in the technical report or appendix.	Opening the executive summary with technical detail.	A senior, non-technical reader with little time.
2	Scoping the problem	Before recommending controls, state scope, constraints (budget, time), and assumptions, including what is out of scope.	Jumping to vendors or trying to fix everything.	A scenario with a budget cap, a deadline, and systems of mixed relevance.
3	Gap analysis	A gap analysis compares the current state to a target state (often a framework) and lists the differences to prioritize.	Listing only the target controls or calling it a compliance certification.	“Current state,” “target,” and “what’s missing.”
4	Risk-based prioritization	Rank findings by risk score = likelihood × impact, not by who raised them or by one factor alone.	Prioritizing by seniority of the requester or by likelihood alone.	Several findings with likelihood and impact ratings.
5	Mapping to NIST CSF functions	Govern sets direction; Identify knows assets and risk; Protect prevents; Detect spots events; Respond acts on incidents; Recover restores.	Mapping monitoring to Protect or restoration to Respond.	A recommendation whose main purpose is to prevent, spot, act, or restore.
6	Total cost of ownership	Compare options on total cost over the same period: upfront cost + recurring cost × years.	Comparing first-year cost only.	An upfront price vs. a subscription over several years.
7	Implementation planning	Roll out in phases — pilot, measure against specific metrics, then expand — with training and communication.	A single big-bang cutover with no metrics.	An approved recommendation affecting thousands of users.
8	Evidence and peer feedback	Support claims with authoritative, current sources and address substantive reviewer feedback before cosmetic changes.	Keeping a weak source because it is popular, or fixing style first.	A reviewer flags a source or claim.

Course scope (public description): Cybersecurity and Information Assurance Capstone is the capstone experience for security and information assurance practitioners. Throughout the program, learners have learned best practices, compliance guidelines, and frameworks necessary in information security. This capstone puts learner knowledge to the test by having them respond to an information security scenario, research and create a proposal, provide an executive summary, and ultimately create a technical report that utilizes feedback from their peers.